ModelIt!

Student Activity Pack

The Nitrogen Crisis: Why Fertilizer Is Destroying Our Rivers

Modeling Nutrient Pollution and Aquatic Ecosystem Collapse

Grade 10L2 | NGSS HS-LS2-4, HS-ESS2-2 | Cause and Effect

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What do you think happens to the fertilizer farmers put on their fields when it rains?

2 Why might too many nutrients in a lake or river be a bad thing — don't nutrients help things grow?

3 Draw a diagram showing how you think nitrogen moves from a farm field to the ocean.

4 What is a 'dead zone' in the ocean and what do you think causes it?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Fertilizer Application | Nitrogen Runoff | Algal Bloom Growth | Oxygen Depletion | Fish Kill Rate | Soil Depletion | Water Treatment Cost

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Conventional Farming

High fertilizer, intensive monoculture

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Precision Agriculture

Optimized fertilizer matched to crop needs

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Soil Regeneration

Cover crops, composting, reduced synthetic input

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Wetland Filtration — The natural capacity of wetlands to absorb and denitrify excess nitrogen before it reaches rivers and oceans — 90% of US wetlands have been destroyed, eliminating this natural buffer

Atmospheric Deposition — Nitrogen oxides from fossil fuel combustion that deposit onto land and water surfaces through rain, adding to nutrient loading independent of agricultural sources

Phosphorus Loading — The companion nutrient that often limits algal growth in freshwater systems — reducing phosphorus may be more effective than reducing nitrogen for preventing freshwater eutrophication

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How does nitrogen fertilizer — designed to help plants grow — end up creating massive dead zones in oceans thousands of miles from the nearest farm? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate the concept of nonpoint source pollution — why is agricultural runoff harder to control than factory discharge?**

2. **What threshold effects did you observe in your model — at what point does nitrogen loading overwhelm the waterway's capacity to process it?**

3. **How does Soil Depletion create a vicious cycle that increases both fertilizer dependence and nitrogen pollution over time?**

4. **What does your model suggest about the economic cost-benefit of preventing pollution at the source versus treating it downstream?**

5. **What are the limitations of modeling a continental-scale nutrient cascade with only seven components when real watersheds involve thousands of farms, tributaries, and microhabitats?**

8 STEM Challenge: Design a Watershed Nutrient Management Plan

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YOUR ENGINEERING MISSION

Design a comprehensive nutrient management strategy for a 5,000-hectare agricultural watershed that maintains crop yields while reducing nitrogen runoff to safe ecological levels.

THE SCENARIO

A farming community downstream from a dead zone event has been ordered by the EPA to reduce nitrogen runoff by 40% within five years or face mandatory fertilizer restrictions. Your team has been hired to design a plan that achieves this reduction while keeping farmers economically viable.

Engineering Design Thinking

1. WHAT IS THE MAXIMUM FERTILIZER APPLICATION RATE THAT MAINTAINS YIELD WHILE MEETING THE 40% RUNOFF REDUCTION TARGET?

2. WHERE WOULD YOU PLACE RIPARIAN BUFFER STRIPS AND CONSTRUCTED WETLANDS TO INTERCEPT RUNOFF MOST EFFECTIVELY?

3. HOW WOULD YOU INCENTIVIZE FARMERS TO ADOPT REGENERATIVE PRACTICES — SUBSIDIES, TECHNICAL ASSISTANCE, OR MARKET PREMIUMS?

4. WHAT DATA WOULD YOU COLLECT TO DEMONSTRATE COMPLIANCE AND MEASURE ECOLOGICAL RECOVERY DOWNSTREAM?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Environmental Scientists study how human activities affect water quality, soil health, and ecosystems. They work for the EPA, state agencies, consulting firms, and agricultural organizations, earning $55,000–$110,000/year. Agricultural Engineers who design sustainable farming systems earn $65,000–$120,000/year.